PURDUE UNIVERSITY
SCHOOL OF CHEMICAL ENGINEERING
ChE 43500 – Chemical Engineering Laboratory SPRING 2026

Lecture: Monday, 4:30 pm, WALC 2007
Laboratory: Section 14 – MW 8:30 – 11:20
Section 02 – MW 11:30 – 2:20

All meet in Forney G014

Instructors: Prof. Enrico Martinez, 496-6998, marti309@purdue.edu
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Prof. Xiaoping Bao, 496-3094, bao@purdue.edu

Course Coordinator: Prof. Enrico N. Martinez,

UPDATES
Check your email, the course website in Brightspace
and the bulletin board outside of G014, regularly, for schedule changes.

Course Objective: Apply chemical engineering fundamentals to design, conduct, and analyze the results of experiments constructed to obtain performance data required to complete design projects.

Course Outcomes: As a result of the activities in this course, students should be able to:
1. Determine the performance data required to complete the design for a process or process equipment.
2. Design and conduct an experimental program to determine the required performance data.
3. Use the requisite performance data to obtain a technically feasible design.
4. Effectively report the results of their work in written form and communicate the results to their peers in an oral presentation.
5. Function as a member of a technically diverse team.
6. Understand different personality types, how these types respond, and how they can function effectively on teams.
7. Develop an understanding of the professional responsibilities of the engineer as a member of a team.
8. Understand and practice appropriate safety measures.

The activities in the course consist of one measurement-analysis project and two design projects, each with nine laboratory sessions. Each team member will either be the team leader, experimental engineer, or design engineer for one of the projects. The course schedule gives the times of each activity for each of the three projects. On the fourth period of each project, you will participate in a planning conference with your instructor. An oral progress report is presented during the eighth period and the final project report is due two periods after the ninth (and final) period of the project. The document, *Notes on the Design Process*, provides more information on the activities of each design project and is available via the class Brightspace Web site.

The first project will be divided into 2 parts:
1. Measurement. The teams will work on the measurement of one or more of the significant variables of the project during the initial 3 sessions of the semester. The results, including statistical data analysis, will be reported in a 2-page memorandum.
2. Analysis. The team members will devote 6 lab sessions to characterize the behavior of their pilot plant equipment and will develop the basic model for the process (reaction kinetics, separation efficiency parameter). The results will be presented in a 5-page report. Details on the writing of the memorandum and report are given in separate documents.

You are expected to act ethically and professionally in this course. You are to submit work that is entirely your own team's. Any evidence of cheating or any other form of dishonesty will result in grade penalties up to a failing grade in the course. You must follow all safety rules. Violation of any of the safety rules may result in penalties up to a failing grade in the course. ± grading will be employed in this course.

The prerequisites for ChE43500 are ChE30600, CHE 32000, CHE 37800 and ChE34800. You must have passing grades in these prerequisite courses to remain enrolled in ChE43500.

Course Staff:
Direction and help are provided by your **instructors**, the **graduate teaching assistants** and **Jeff McGlothlin** (Laboratory Manager). The **instructor** fills the role of an industrial manager concerned with evaluating (1) your plans, (2) the technical content of your oral and written reports, and (3) the soundness of your conclusions. The **communication specialist** will evaluate (1) the presentation skill of your oral reports and (2) the non-technical aspects of writing in your reports. **Mr. McGlothlin** is primarily responsible for the laboratory equipment. He will instruct you on how to operate the equipment efficiently and safely. Be sure to talk to him before operating the equipment and see him if there are equipment difficulties.

SECTION	SUPERVISOR	TEACHING ASSISTANT	LAB G014	CONSULTATION
02	E. Martinez	Emily Czaplewski	MW 11:30-2:20	Mondays 3:30-4:20
14	K. Thomson	Xihui Zhao	MW 8:30-11:20	TBA by Instructor

LABORATORY MANAGER	Jeff McGlothlin	jgloth1@ecn.purdue.edu
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TEACHING ASSISTANT (SEC. 14)	Xihui Zhao	zhao1089@purdue.edu
COURSE COORDINATOR	Enrico Martinez	marti309@purdue.edu

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1.0 Goals and Methods

Your goals in ChE 435 are to learn to:

- make technically sound measurements and evaluate data,
- make realistic plans for experiments,
- compare statistically significant results with theory as well as with literature,
- sharpen skills in solving problems,
- manage a project and work in a team environment,
- design a process according to the requirements stated, and
- communicate effectively in oral and written reports.

1.1 Attendance, Excused Absences and Grading Policy

Attendance at all class meetings and laboratory sessions is mandatory. Absences will result in penalties and excessive absences may be reported to the Dean of Students for appropriate action. Interviews and plant trips may be considered "excused absences" as long as you notify the instructor at least one week prior to the event.

- Attend all lectures and laboratory sessions.
- Report plant trips at least one week in advance and document illnesses with your Instructor and TA.
- Make up any excused omission of work in coordination with your team and instructor.
- **An unexcused absence in a lecture may result in loss of 20 points.**
- **An unexcused absence in the laboratory may result in loss of 50 points.**
- **Failure to attend a scheduled appointment with course staff may result in the loss of 10 points.**

1.2.1 **Course Textbook:** ***Transport Processes and Separations****,* 5th ed., Geankoplis Ch. J., Hersel, A. A., and Lepek, D. H., Prentice Hall, 2018

Major References:

[1] ***Unit Operations of Chemical Engineering*** 7th ed., McCabe, W. L., J. C. Smith and P. Harriott, McGraw-Hill, 2005

[2] Denn, M. M., ***Chemical Engineering: An Introduction****,* Cambridge University Press, 2012.

[3] ***Chemical Engineers' Handbook*** 7th ed., R.H. Perry & D.W. Green (eds.), McGraw-Hill, 1999

[4] ***ChE 435: Chemical Engineering Laboratory***, Library website, http://guides.lib.purdue.edu/che435, developed by Jeremy R. Garritano, Spring 2012, updated by Dave Zwicky 2020.

Recommended reference: (available in the laboratory & from Communications Specialist):

[1] ***Handbook of Technical Writing****,* 7th ed., Brusaw, C.T., G.J. Alred, and W. E. Oliu (eds.), St. Martin's Press, New York, 2003.

2.0 Industrial Simulation

You are the chemical engineer and your instructor will be a manager who will give you problem assignments with some specifications that may be vague or incomplete. Your task is to:

- formulate the problem in detail and understand the issues involved;
- design a set of experiments to plan around limitations of time, equipment and personnel;
- evaluate results, possibly change plans during the project and draw appropriate conclusions;
- present an oral progress report and deliver a final written report to management.

2.1 Details

Use laboratory-supplied data sheets to record data **in ink**, **no pencil**. Be clear and neat so that your instructor can do the calculations independently (identify numbers and give the units). At the end of the period, the assistant will witness your data sheets if your work area is clean, equipment has been returned, and sample bottles are labeled and stored properly. Take the original with you and leave a copy with the assistant. All original data sheets must be attached to your final report.

2.2 Teamwork

Three persons will work together as a team for each project. There are three roles on the teams: leader, experimental engineer, and analyst (see also Table 1 and Section 3). You will perform one role for a project and then rotate to another role for the next project. Each member shares responsibility for the technical and written quality of the entire report.

2.3 Projects

Some of these projects draw on material from your courses, but many projects will include topics that are not yet familiar to you. You are expected to augment your knowledge with textbooks, journal articles, and visits to the library or the Internet.

3.0 Individual and Team Roles for Analysis and Design Projects

This section describes each team member's role and responsibilities during projects and when writing reports.

3.0.1 Team Leader

As Team Leader you:

- determine the overall direction of the project, set the objectives, and schedule the activities of the team,
- direct the team to do experiments and calculations,
- meet the deadlines of the project, and
- plan the written report, and present the oral progress report.

Table 1: Team Leader Sections

Analysis Project	Design Project
Cover Page	Cover Page
Abstract	Abstract
Table of Contents	Table of Contents
Introduction	Introduction
Theory/Method	Theory/Method
Notation	Notation
References	References
	Derivations

3.0.2 Experimental Engineer

As the Experimental Engineer you:

- become familiar with the safe operation of the apparatus and analytical chemistry,
- formulate a design of experiments with factors varied and levels selected,
- communicate with the Design Engineer to be sure the team gets the necessary data that will be used in the calculations, and
- *perform a safety review before operating the equipment!*

Table 2: Experimental Engineer Sections	
Analysis Project	**Design Project**
Apparatus	Apparatus
Procedure	Procedure
Data Sheets	Design of Experiment
Calibrations	Results
	Discussion
	Data Sheets
	Experimental Sample Calculations
	Intermediate Number Tables
	Calibration Curves

3.0.3 Analyst/Design Engineer

As Design Engineer you:

- collaborate with the experimental engineer and leader on the data to acquire,
- analyze and interpret the data and develop the conclusions,
- apply statistics (regression, analysis of variance, and propagation of error) to the data,
- use these results and the equations described in the Theory/Method section to actually design the equipment and the process that you recommend to carry out the task that you have been given in the project handout.

Table 3: Analyst/Design Engineer Sections	
Analysis Project (Analyst)	**Design Project (Design Engineer)**
Results	Design Calculations
Discussion	Final Design
Conclusion & Recommendations	Recommendations
Sample Calculations	Design Sample Calculations
Calculation Tables	Initial Design Tables
	Process Flowsheet

4.0 Events

Along with your laboratory work there will be lectures, planning conferences, and oral and written reports. Teamwork, which is integral to industrial practice, is stressed throughout the course. There is no final exam for the course.

Figures 1 and 2 show ways of scheduling your activities for maximum effectiveness--do "parallel processing."

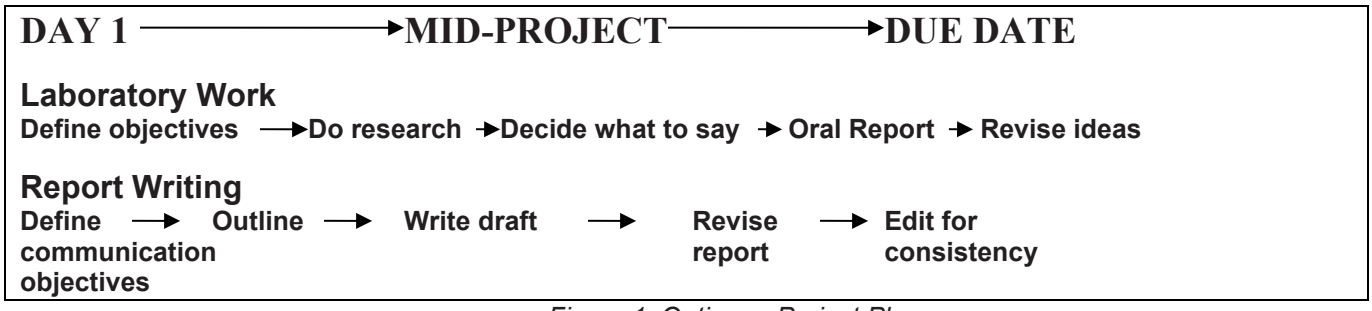


Figure 1. Optimum Project Plan

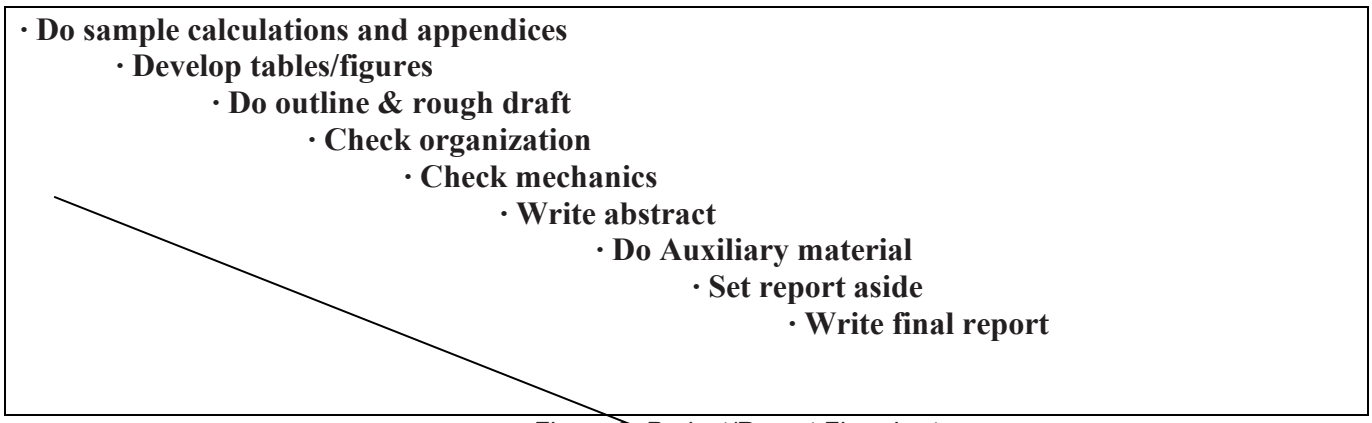


Figure 2. Project/Report Flowchart

4.1 Design Projects - Planning Conference

During the fourth period of each project, teams meet with the **instructor** to present plans. Each person prepares an organized 5-minute discussion. Some instructors may meet with you informally in the laboratory; others may require you to prepare a Power Point presentation.

Leader: Presents objectives, plans, theory/ideas/principles, and a brief timetable.

Experimental Engineer: Describes the key aspects of the apparatus as related to principles and shows the preliminary technical and statistical designs of the experiments.

Design Engineer: Presents some use of the theory, methods of calculating results, how the statistical calculations will be done and what results or conclusions are expected as well as handing in the information audit.

To prepare for the conference, read the assignment sheet, read background material, and talk to the **instructors**, the **graduate teaching assistants,** and **Rick McGlothlin** about the apparatus.

4.2 Oral Progress Reports

On the **eighth period of each project**, the **team leader** gives a **10-minute** oral progress report on work to date plus directions for completing the project and for future work. The talk is videotaped for later review with the communication specialist.

During the presentation, the speaker responds to questions from the audience to defend the approach and the results. The instructor grades for technical merit and may have questions or suggestions about the work. The communications specialist grades the presentation.

Clearly, the explanation of your results will be the most important for your professional audience, so don't use up most of the time describing the apparatus etc. By the time of the progress report, you must plan to have a reasonably balanced product, including key results and conclusions. You will use the remaining laboratory periods to complete the work. See Figures 1 and 2 for ways to optimize time and effort.

4.2.1 Oral Report Specifications

Length

10 minutes - One point will be subtracted for each minute over the allotted time. Points may be deducted for presentations deemed too short.

What to wear

Business casual or better

Grading Criteria

A copy of the communications assessment is available via Blackboard in the rubrics section. In general the communications assessment is based on:

- *Organization & Style* – Does the flow of your presentation make sense? Did you include the appropriate information?
- *Visual Aids* – Slides should be easy to read and serve as an outline of your presentation. Formatting should be consistent throughout the slides.
- *Presentation techniques* – Volume, pace, eye contact, movement and presentation style will be evaluated.

- *Question & Answer* – Encourage questions from audience and answer them as best you can. Ask your teammates for their expertise or to weigh on the question as needed.

Each presenter is required to meet with the communication specialist following his/her presentation to evaluate their performance. Failure to do so will result in a 10 point deduction. Information about performance review appointments will be provided during the semester. Any presenter who is unprepared to present on the assigned date will receive no more than half credit from the communications specialist.

Information to present – Analysis Project

- Introduce self, team, project, supervisor (use title slide)
- Outline the presentation
- State the objectives
- Preview the results
- Explain as much of theory involved as audience needs to know
- Describe the apparatus and procedure
- Discuss results to date
- Give directions for future work

Information to present – Design Project

- Introduce self, team, project, supervisor (use title slide)
- Outline the presentation
- Discuss relevance in real world, applications, etc.
- State the objectives
- Preview the results
- Explain as much of theory involved as audience needs to know
- Briefly describe the apparatus and procedure
- Discuss results to date
- Present final design & recommendations
- Give directions for future work

4.3 Written Reports

*Teams are responsible for submitting **TWO** copies of written reports at the **BEGINNING** of the designated laboratory period. One copy should include the appendices; the other is to be the complete report without the appendices. Both copies should include a project cover sheet.*

Your supervisor will evaluate the copy with appendices. The communications specialist will evaluate the copy without the appendices.

Even tEven though each team member writes assigned sections of each report, the finished report should read as a smooth, continuous, coordinated document. Your **instructor** will assess organization and technical content, including (1) your experimental design, analysis of error, (2) the conclusions you draw from internal checks (mass and energy balances, duplicate runs, and comparison of propagated error with regression standard deviation), and (3) comparisons with literature or theory. **The communication specialist** will evaluate your writing.

You should expect to receive feedback on the technical content of your report within two weeks of submission and on the technical communications within three weeks of submission.

The following tables show the writing responsibility for each person in the team as well as the order of the sections that comprise the report.

Table 4. The Analysis Report
Responsibility for and Order of Sections

Leader (L)	Experimental Engineer (E)	Analyst (A)
Cover Page		
Abstract		
Table of Contents		
Introduction		
Theory/Method		
	Apparatus	
	Procedure	
		Results
		Discussion
		Conclusion & Recommendations
Notation		
References		
	Data Sheets	
		Sample Calculations
		Calculation Tables
	Calibrations	

Table 5. The Design Report
Responsibility for and Order of Sections

Team Leader	Experimental Engineer	Design Engineer
Cover Page		
Abstract		
Table of Contents		
Introduction		
Theory/Method		
	Apparatus	
	Procedure	
	Design of Exp.	
	Results	
	Discussion	
		Design Calculations
		Final Design
		Process Flowsheet
Notation		Recommendation
References		
	Data Sheets	
	Experimental Sample Calc.	
		Design Sample Calc
		Int. Design Tables
	Calibration curves	
Derivations		

Remember: The order of the sections in the report follows the order in the table. 4.3.1

4.3.1 Report Section Descriptions

Cover Page
The cover page must include the title of the project, the course name and number, your section number, and the names and roles of your team members.

Abstract
An abstract answers what-why-how questions. It is a mini-report that has to be able to stand alone, independent of the rest of the report. Give objectives and results. Be specific, include only final equations (if relevant to the overall objective) and key numbers, and clearly state the main conclusions. Tables and figures should not be included. The abstract should not exceed 250 words.

Table of Contents
Include all the sections. **Remember to list the figures and tables, complete with captions.** Note that graphs, schematics, illustrations, photographs, etcetera are all labeled as "figures."

Introduction
Set the real-world context/relevance of the project and list the objectives. Give a short preview of the results of the report.

Theory/Method
Briefly state and explain the ideas, principles and techniques, and the relevant equations (including assumptions and references) that apply to your project. Define each symbol the first time it is shown. Don't simply present a list of equations (which makes the reader do your work for you—and that makes for an unhappy reader). For example, if the objective is to find a heat-transfer coefficient, show first the equation that defines the heat-transfer coefficient (remember to state the character of the equations you use—definition, mass or energy balance, kinetic expression, correlation). Then show the equation that gave you the rate of heat transfer. (See Section 4.2)

Apparatus
Describe the main features of the apparatus in words and with a schematic or figure: a page with a schematic alone is **not** enough to constitute a description of the apparatus. Be careful in drawing the schematic. Note that a "schematic" has specific requirements and is not meant to be a "picture" or photographic reproduction. See the *Handbook of Technical Writing* section on "schematic diagrams."

Procedure
Narrate what you did. Don't just copy the procedure given in the handout, that is, don't use a cookbook approach that tells someone else how to do the procedure. Point out controlled and measured variables. Discuss sensitivity, accuracy, and processing issues. Describe the design used and the experiments that were actually done.

Results and Discussion
Discuss and interpret your results, using plots (figures) and tables. Describe what is shown in the figures and tables and tell what it means. Don't make the reader figure it out. For tables and figures in the main body, describe each in a caption and refer to each in the text. Put long, extended tables in the Appendix; show examples of some plots and then discuss the general results. Present results of your statistical work, including regression, propagation of error and analysis of variance. Refer to the Appendix for details.

Conclusions and Recommendations

Your readers want to know what makes the information in your report significant. The conclusions answer this concern. Present the following items factually and in a positive style: Don't be negative about your work.

- Start with an introductory sentence; then put the conclusions in a numbered list. Answer the objectives.
- Relate them to the objectives presented in the introduction and then add the recommendations that are supported by the results of your project.
- Present no new material in the conclusions. Each conclusion must be summary statement of evidence presented in the results and discussions sections.
- Make your recommendations the logical outcomes of the conclusions. Phrase the statements as "it is recommended that…"
- Note whether internal checks such as material and energy balances close (as shown by a "t test"), and if not, what is the most probable reason for this and what might be done about it in future work. Don't supply a laundry list of everything that might account for it. Be specific.
- Compare your results to literature results and determine if yours are significantly different; and if so, tell why and what that means for future work. For example: "The Nusselt number of the heat exchangers is always higher than the literature values because the literature work had a 50-pipe-diameters calming section before the test section. There is a right-angle bend at the entrance of the water, which creates extra turbulence. Exchangers with entrance elbows will be researched in the future."

Notation

List symbols used in alphabetical order (English first and Greek in a second list). (But list them also, in an indented table, when they are first mentioned.) Always include units.

References

List books and articles referred to in the report. (See Section 4.3)

Appendices (Derivations)

Appendices contain details needed to support your results and to be used by people who may continue your work. Make it easy for them by organizing the material under subtitles or sub-appendices listed in the table of contents. Use enough words in appendices to define units and manipulations, but don't worry about smooth text with introductions and transitions.

Appendices (Data Sheets, Calibrations)

Appendices contain details needed to support your results and to be used by people who may continue your work. Make it easy for them by organizing the material under subtitles or sub-appendices listed in the table of contents. Use enough words in appendices to define units and manipulations, but don't worry about smooth text with introductions and transitions. Include data sheets, sample calculations (equations, numbers, units) including error propagation, tables of intermediate variables in the calculations, working plots, calibration curves, and all the figures and tables that you did not include in the main body.

Appendices (Sample Calculations, Calculation Tables)

Appendices contain details needed to support your results and to be used by people who may continue your work. Make it easy for them by organizing the material under subtitles or sub-appendices listed in the table of contents. Use enough words in appendices to define units and manipulations, but don't worry about smooth text with introductions and transitions. Include data sheets, sample calculations with error propagation, tables of intermediate variables in the calculations, working plots, calibration curves, and all the figures and tables that you did not include in the main body.

5.0 Related Format Guidelines

Every organization has its own way of doing things—conventions or styles. The following are some format matters that our "company" insists on. The main guidelines to follow are in the *Handbook of Technical Writing (Alred et al. 2003)*.

5.0.1 Figures and Tables

Graphs are almost always the preferred method of presenting results. They must clearly show data points and curves for the model developed. Label axes and include units. Label all plots, drawings, schematics and photographs as "Figures." Unless you can integrate them smoothly into the running text, put tables and figures on separate sheets of paper. Number both tables and figures consecutively in the order of their appearance in the report, beginning with "1."

Referring to Tables and Figures

Help the reader by placing the table or figure close to where you refer to it; however, very detailed data tables should be put in the appendix. Call attention in the text to trends or other features shown in the table or figure that are important to your development; never assume it is obvious to the reader.

For example:

"In Figure 3 on page 8, the Nusselt number is plotted against the Reynolds number for the four exchangers. The prediction of the Dittus-Boelter equation is also shown. For each exchanger, Nusselt number increases with Reynolds number at a rate not significantly different from that of the prediction, but the 95% confidence intervals show that Nusselt numbers for all exchangers are significantly different from the values predicted by Dittus-Boelter."

Labeling Tables & Figures:

(Graphs/plots, schematics, photographs, etc. are all called "Figures." Start with number 1 and label them consecutively throughout the report. Ditto for tables.)

Tables are labeled at the top.

Table 14. Heat-transfer Coefficients (3 in. ID Exchanger)
Table labels: Above the table.

Figure labels: Below the figure.
Figure 7. Heat-transfer Coefficient as a Function of Reynolds Number. Runs made after replacing the pressure gage. Ð, 0.5-in. tube; —, 1-in. tube; Ð, 3-in. tube.

Figures are labeled at the bottom.

Figure 3. Labeling Tables and Figures

5.0.2 Equations

Equations must be set in context and explained. Don't make the busy reader do your work for you. State their nature or origin (definition, balance, kinetic expression, or empirical correlation) to take the mystery out of your development. Guide the reader through your thinking. Here's an example from the theory section of a report:

The "1" tells us, this is the first reference cited in the report and that the equation is on page 668. Hereafter, every reference to this text will still be "1" with the appropriate page number.

The calculation of the energy transferred to the cooling water is shown in the following relationship (1, p. 668):

$$\Delta H_{cw} = m_{cw} C_p (T_{co} - T_{ci}) \qquad (5)$$

where m_{cw} = Mass flowrate [lb_m/min]
C_p = Specific heat [BTU/lb_m/°F]
T_{co} = Temperature of outlet stream [°F]
T_{ci} = Temperature of inlet stream [°F]

Remember to include the notation here for each equation and also on the notation page.

Figure 4. Sample Equation

This is Equation (5) taken from the reference (1, p. 668). See Sec. 4.3 for more about citing literature. Include notation here, with units, as well as in the notation section.

5.0.3 References to the Literature

The ideas, equations, schematics and other general background material that you'll use will come from a variety of sources that you absolutely must acknowledge. These sources will either be from books which are written by the authors whose names appear on the title page, or books which are collections of articles by editors who assemble them for publication, or from journals or magazines, or even various sources ranging from brochures to phone conversations to locations on the internet. Here's how to acknowledge them in text:

"The calculation of the energy taken into the cooling water is shown in the following relationship (1, p. 668)."

In the reference section this reference is the first one cited (that's why it's number 1!) and it just happens to be the following one:

> [1] Greenkorn, R., and D. Kessler, *Basics of Momentum, Heat, and Mass Transfer,* Marcel Dekker, New York, 1999.

If this text is referenced again, only the page number will change, for example, (1, p. 297). Even if you refer to it 20 times, it will still be 1, only the pages numbers will/might change. The next text or source that you cite will be "2."

The kinds of texts/sources you might encounter:

- **Authored Text -** Author(s) names go first.

Bird, R.B., W.E. Stewart, and E.N. Lightfoot, *Transport Phenomena*, Wiley, New York, 1960.

- **Edited Text -** Title first, then editors' names.

Chemical Engineers' Handbook, 6th ed., R.H. Perry, D.W. Green, and J.O. Maloney (eds.), McGraw-Hill, New York, 1984.

- **Journal Article -** Some omit titles. Follow the usage of the journal you are citing.

Kramers, H., and P.J. Kreyger, "Title of Article," *Chem. Eng. Sci.*, ***6*** 42 (1956).

• Laboratory Report –

Warf, J.A. and M. Tucker, Design of a Column to Scrub Nylon Plant Vent Gas, Ch.E. Laboratory Report, Purdue U., 20 March 1988.

• Phone conversation

Telephone conversation with J.A. Brown, CEO, Megastar Industries, July 24, 1998.

• Website

Purdue University School of Chemical Engineering. (2014). *Title of Page.* [Online] Available: http://www.ecn.purdue.edu/Engr/.

• Other Miscellaneous Sources – See a copy of the Handbook of Technical Writing in the laboratory or in the office of the communication specialist for assistance with citing sources that aren't listed in this publication.

Be helpful. Include any information you as a reader would want to know that would enable you to locate the source.

5.0.4 Additional Information

- Use past tense when writing reports for this course. In certain instances, such as when describing theory, it may be necessary to use present tense.
- NEVER use I, we, our or us when writing a technical report. Instead reference the team, the experimental engineer, the team leader, etc.
- Do not use passive voice when writing for this course. More information about passive vs. active voice is available in the *Handbook of Technical Writing*.

A grammar heads-up: "Who's doin' what to who?" [**Dangling modifiers**]

Think about what you're writing/saying. Each sentence typifies problems in modification, especially dangling modifiers (see the Handbook).

- The rabbit was seen using binoculars. [*that's how it kept track of the fox!*]

- The car was designed by the safety engineer with an impact-absorbing rear end. [*no comment!*]

- After leaving the laboratory, the parameters were checked. [*did they leave single-file?*]

- A material balance was solved around each reactor applying Equation 1. *[smart reactor!]*

- Samples of the new composite were sent to NASA to study in the new plant. [*studious samples!*]

Always make sure there is a logical doer of the action in the sentence itself.